

Executive Summary

This covers four stages data analysis: data cleaning, population analysis, trend analysis, and statistical testing. The first project focused on preparing a household survey dataset for analysis. Inconsistent entries, blank rows, duplicate household records, missing ages, and different date formats were identified and corrected. After cleaning, the dataset contained 1500 unique household records, with missing ages handled using state-level median values.

The second project focused on population analysis by state, sex, age group, and place of residence. Population totals, urban and rural shares, dependency ratios, and a population pyramid were used to turn data into information suitable for reporting and dashboard use.

The third project examined changes in literacy, fertility, under-five mortality, and primary-school enrolment between 2015 and 2023. The results showed general improvements across the states, although the pace of progress differed. The analysis also showed a strong negative relationship between literacy and under-five mortality.

The final project used statistical tests to examine income and other household characteristics. Monthly income was strongly uneven, making the median more representative than the mean. There was no significant income difference between male and female respondents, no significant relationship between education and employment status, and education level alone explained very little of the differences in monthly income.